柯阝然. 未央-水木22 工程经济学

1. ~~NAV~~ 等额年值法：

~~NAV~~$_A$=

A：(150+15×(P/A，ic，15))×(A/P，ic，15)

= 34.72

B = (100+10×(P/A，ic，10))×(A/P，ic，10)

= 36.27454

A更佳

最小公倍数法：

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	150	15	15	15	15	15	15	15	15	15	15	15	15	15	15
B	100	20	20	20	20	20	20	20	20	20	20+100	20	20	20	20

	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	15+150	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
B	20	20	20	20	20	20+100	20	20	20	20	20	20	20	20	20	20

A：327.3125 B=341.957 A更佳

2.

	0	1	2	3	4	5	~~6~~	~~7~~	~~8~~	~~9~~	~~10~~	~~11~~	~~12~~	~~13~~	~~14~~	~~15~~
A	-300	96	96	96	96	96~~-300~~	~~96~~	~~96~~	~~96~~	~~96~~	~~96-300~~	~~96~~	~~96~~	~~96~~	~~96~~	~~96~~
B	-100	42	42	~~42-100~~	~~42~~	~~42~~	~~42-100~~	~~42~~	~~42~~	~~42-100~~	~~42~~	~~42~~	~~42-100~~	~~42~~	~~42~~	~~42~~
Δ	-200	54	54	54 ~~154~~	96 ~~54~~	96 ~~-246~~	~~154~~	~~54~~	~~54~~	~~154~~	~~-246~~	~~54~~	~~154~~	~~54~~	~~54~~	~~54~~

ΔIRR = ~~21%~~ 19.7% >10% 选A

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3. 利用Excel计算NPV

残值相等。

$NPV_1 = 2880.429$

$NPV_2 = 2878.318$

$NPV_3 = 2888.659$

选择C

4.

1) $C_F = \frac{4140}{6} = 690$万元

~~$\frac{690\times10^4}{1.8\times10^4\times(5500-3200)}$~~

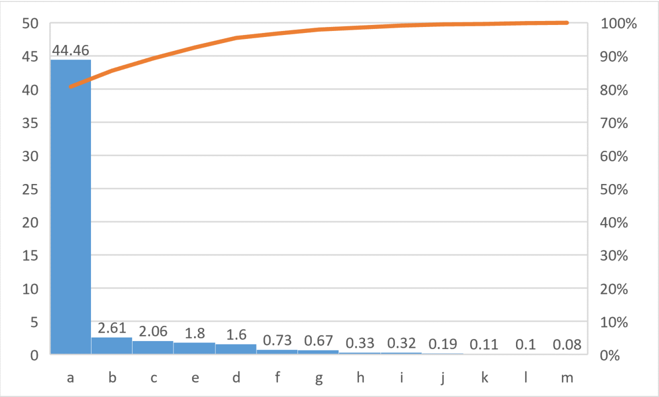
$$BEP(\%) = \frac{690\times10^4}{1.8\times10^4\times(5500-3200)} = 16.7\%$$

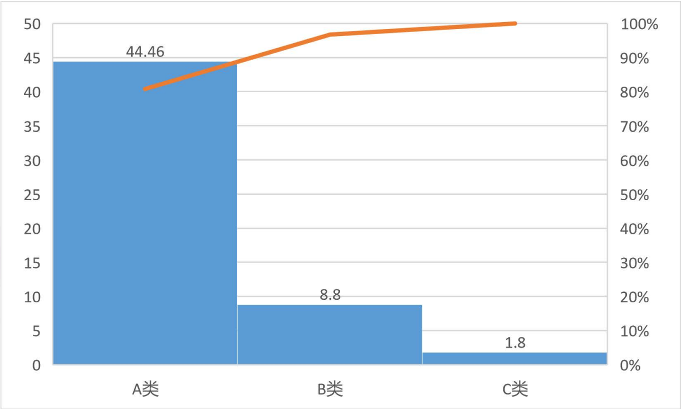
2) ~~价格~~

	-10%	0	+10%
价格	21.9%	16.7%	13.5%
固定成本	15%	16.7%	18.3%
变动成本	14.6%	16.7%	19.4%

价格、变动成本

5. 见下　A类：a　C类：e　其余，B类





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6、

	A	B	C	D	E	F	G		功能评价系数	成本系数
A		1	1	1	0	1	1	5	5/21	0.152
B	0		1	0	0	1	0	2	2/21	0.127
C	0	0		0	0	1	0	1	1/21	0.076
D	0	1	1		0	1	1	4	4/21	0.203
E	1	1	1	1		1	1	6	6/21	0.291
F	0	0	0	0	0		0	0	0	0.051
G	0	1	1	0	0	1		3	3/21	0.101

价值系数

1.567
0.752
0.627
0.940
0.981
0
1.411

选A

7. ① 8% 36,480,000
15% 68,400,000

1997 30,880,000

Goldman, Sachs & Co.	~~18,528,000~~ 19.5×10⁶
David Rockefeller	~~4,632,000~~ 4.88×10⁶
Gianni Agnelli	~~3,088,000~~ 3.25×10⁶
Stavros Niarchos	~~3,088,000~~ 3.25×10⁶
Tishman Speyer Properties, L.P.	0

1998 40,960,000

Goldman, Sachs & Co.	~~23,040,000~~ 24.576×10⁶
David Rockefeller	~~5,760,000~~ 6.144×10⁶
Gianni Agnelli	~~3,840,000~~ 4.096×10⁶
Stavros Niarchos	~~3,840,000~~ 4.096×10⁶
Tishman Speyer Properties, L.P.	~~1,480,000~~ 2.048×10⁶

1999同1998

未央实践

扫描全能王 创建

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7. ② $19.5\times10^6\times(1+IRR_G)^{-1}+\underset{24.576}{\cancel{6.}}\times10^6\,[(1+IRR_G)^{-2}+(1+IRR_G)^{-3}]+540\times10^6(1+IRR_G)^{-3}=288\times10^6$

$IRR_G=29.8\%$

③ 1. 增加公司对于产品销售和供应商的控制力度
2. 减少不确定性因素
3. 获得更多关于市场和供应商信息.

物业管理 $120\times10^6\times2\%=2.4\times10^6$
物业权益 $540\times10^6\times5\%\div60\%=45\times10^6$
建造收益 $100\times10^6\times5\%=5\times10^6$

$2.4\times10^6\times(1+IRR_{TSP})^{-1}+9.448\times10^6\times(1+IRR_{TSP})^{-2}+49.448\times10^6(1+IRR_{TSP})^{-3}=24\times10^6$

$IRR_{TSP}=41.2\%$

④ 租金小于2亿，TSP无分红

亏损 4.096×10^6 美元

8. (1) 土地费用 95340 6810 74860 900
方案a
IRR = ~~9%~~ 7%
投资 1107135000
NPV_A = ~~-38826145~~ -118424001.8

具体数据见后

方案b
IRR = 11%
投资 +729330000
NPV_B = 26450143.81

(2) 提升整体形象与档次
带动赢利
方案a 写字楼之间竞争，且影响口碑
西塔自身：写字楼

(3) ①可提升天河区影响力
②丰富产业结构
③提升总部经济效应
④促进创新创业

①物业管理服务
②智能化出入门禁系统等

未央实践

扫描全能王 创建

	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
土地费	-119935000																													
贷款		-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-28512000	-478512000
投资估算	-277991666.7	-277991667	-277991666.7	-153225000																										
出租费用						141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720	141048720
运营费用						-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744	-28209744
售卖费用		411730000	274486666.7	137243333.3																										
停车位租金				12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000
折旧	-32906666.67	-32906666.67	-32906666.67	-32906666.67	-32906667	-32906666.67	-32906666.67	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906667	-32906666.7
经营税金及附	0	-22645150	-15096766.67	-8261183.333	-712800	-8470479.6	-8470479.6	-8470479.6	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470480	-8470479.6
房地产税	0	0	0	0	0	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360
土地增值税		-8343147	-5562098	-2781049																										
年利润	-397926666.7	74238036.33	-52675864.67	-42575899	-16264800	83668136.4	83668136.4	83668136.4	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	83668136	-366331864
所得税	0	-24498552	0	0	0	-27610485.01	-27610485.01	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	-27610485	0
	-397926666.7	49739484.34	-52675864.67	-42575899	-16264800	56057651.39	56057651.39	56057651.4	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	56057651	-366331864
	100%	1.1	1.21	1.331	1.4641	1.61051	1.771561	1.9487171	2.1435888	2.3579477	2.5937425	2.8531167	3.1384284	3.4522712	3.7974983	4.1772482	4.594973	5.0544703	5.5599173	6.115909	6.7274999	7.4002499	8.1402749	8.9543024	9.8497327	10.834706	11.918177	13.109994	14.420994	15.863093
0.1	-397926666.7	45217713.04	-43533772.45	-31987903.08	-11109077	34807391.07	31643082.79	28766438.9	26151308	23773916	21612651	19647865	17861695	16237905	14761732	13419756	12199778	11090707	10082461	9165873.9	8332612.7	7575102.4	6886456.8	6260415.2	5691286.6	5173896.9	4703542.6	4275947.8	3887225.3	-23093344
NPV	-118424001.8																													
IRR	7%	注：没有按揭折旧算																												
	-1107135000																													

	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
土地费	-119935000																													
贷款		-28512000	-28512000	-28512000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-79200000	-1329200000
投资估算	-153225000	-153225000	-153225000	-153225000	-149720000	-149720000	-149720000																							
出租费用						220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120	220317120
运营费用						-44063424	-44063424	-146555424	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-1.47E+08	-146555424
酒店收入								157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000	157680000
停车位租金				12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000	12960000
折旧	-39399833.33	-39399833.33	-39399833.33	-39399833.33	-39399833	-39399833.33	-39399833.33	-39399833.3	-39399833.3	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833	-39399833.33
经营税金及附	0	0	0	-712800	-712800	-12830241.6	-12830241.6	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502642	-21502641.6
房地产税	0	0	0	0	0	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360	-5148360
土地增值税		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
年利润	-39399833.33	-67911833.33	-67911833.33	-55664633.33	-106352633	52635261.07	52635261.07	99150861.1	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	99150861	-1150849139
所得税	0	0	0	0	0	-17369636.15	-17369636.15	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	-32719784	0
	-39399833.33	-67911833.33	-67911833.33	-55664633.33	-106352633	35265624.91	35265624.91	66431076.9	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	66431077	-1150849139
	100%	1.1	1.21	1.331	1.4641	1.61051	1.771561	1.9487171	2.1435888	2.3579477	2.5937425	2.8531167	3.1384284	3.4522712	3.7974983	4.1772482	4.594973	5.0544703	5.5599173	6.115909	6.7274999	7.4002499	8.1402749	8.9543024	9.8497327	10.834706	11.918177	13.109994	14.420994	15.863093
0.1	-39399833.33	-61738030.3	-56125482.09	-41821662.91	-72640280	21897178.48	19906525.89	34089646.4	30990588	28173262	25612056	23283687	21166988	19242717	17493379	15903072	14457338	13143034	11948213	10862012	9874556.3	8976869.4	8160790.3	7418900.3	6744454.8	6131322.6	5573929.6	5067208.7	4606553.4	-72548849.1
NPV	26450143.81																													
IRR	11%	注：按揭折旧计算																												
	-729330000																													